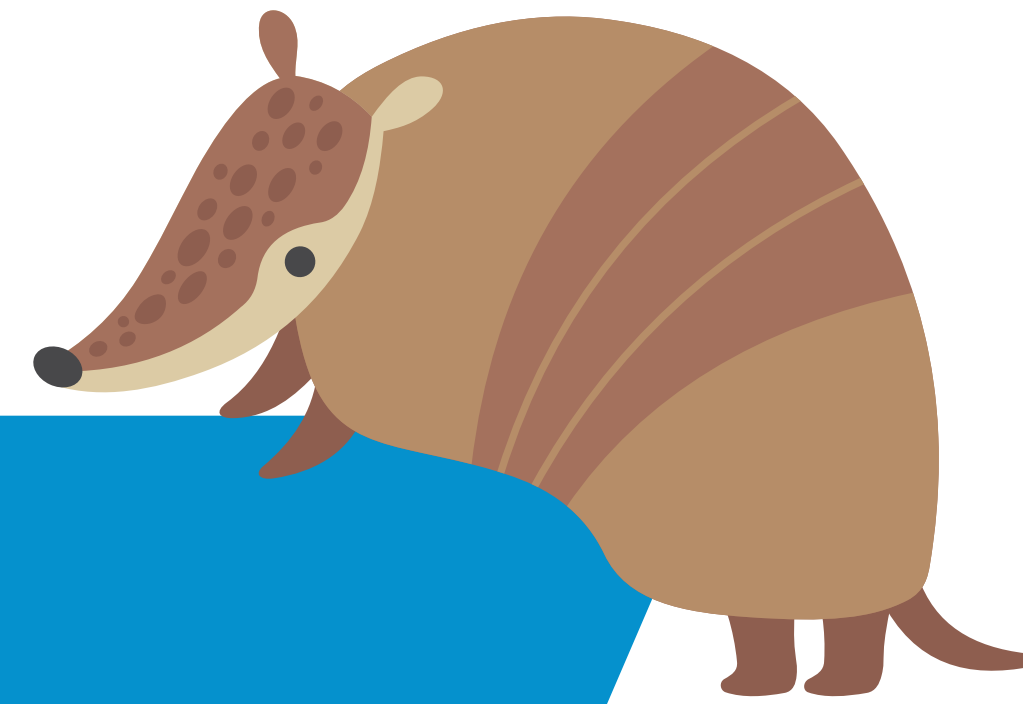




# Automating Filtered Data Extension Refreshes in Marketing Cloud Engagement Using APIs



Presentation By Fnu Avi

# OUR SPEAKER

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**Fnu Avi**

Senior Software Engineer, University of Southern California (USC)

<https://www.linkedin.com/in/fnu-avi/>

# Session Agenda & Goals

## 1. Problem & Context (5 Min)

Establishing the limitation of Filtered Data Extensions (FDEs) and assessing current native workarounds.

We'll examine why native setups introduce overhead and lay down the framework for REST API execution.

## 2. Live Coding Demo (15 Min)

Establishing the limitation of Filtered Data Extensions (FDEs) and assessing current native workarounds.

We'll examine why native setups introduce overhead and lay down the framework for REST API execution.

# The Filtered DE Challenge



## The Data Stagnation

Unlike standard SQL segments, Filtered DEs are inherently static and do not automatically dynamic-sync unless directly prompted.



## Manual & UI Drag

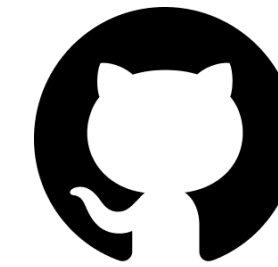
Refreshing requires manual clicking inside Email Studio or tedious provisioning of individual Filter Activities in Automation Studio.



## The API Opportunity

An undocumented, fully functional REST API route bypasses the UI and initiates instantaneous Segment Rebuilds on-demand.

# The 3-Step API Automation Sequence



1

## Get Token

Exchange secure Client Credentials with tenant auth endpoint for access\_token authorization.

```
# Request OAuth2 Context
POST /v2/token
auth.marketingcloudapis.com
```

2

## Get DE ObjectID

Locate the internal customobjectid using the target DE's External Customer Key query.

```
# Query Target Metadata
GET /rowset
customobjectdata/key/{{Key}}
```

3

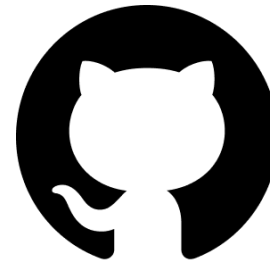
## Refresh Filtered DE

Instruct target engine to immediately rebuild the segmentation filters asynchronously.

```
# Programmatic Run
POST /refresh
filteredCustomObjects/{{ID}}/refresh
```



# Deploying Code in SFMC



To run this within SFMC CloudPages or Script Activities, we consolidate the REST calls into a native SSJS code block.

- **Dynamic Loop:** Supports batch array processes to trigger multiple FDEs simultaneously.
- **Token Caching:** Pulls token context dynamically before executing looping structures.
- **Self-Contained:** Implements zero exterior stack dependencies.

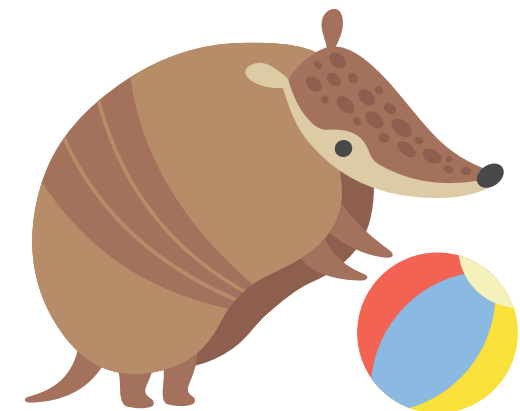
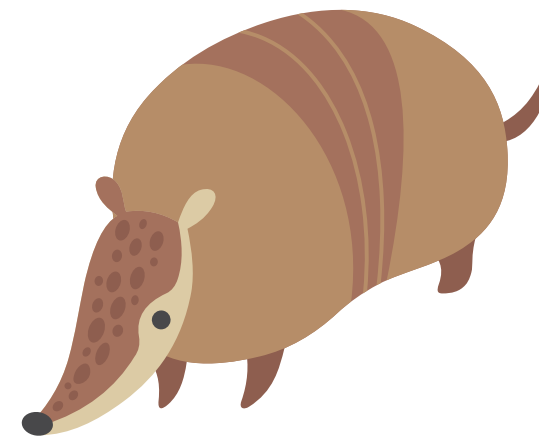
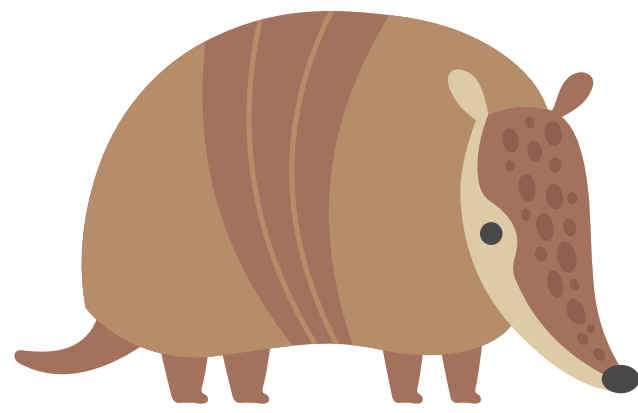
```
MC_TokenRefreshFilteredDE.ssjs

<script runat="server">
    Platform.Load("core", "1");
    var objectKeys = [
        "67ae2a16-f376-f011-ba7a-5cba2c198cc0", // OR Providers
        "68c696f3-3a51-f011-ba7a-5cba2c198cc0" // WA Providers
    ];

    // Call-1: Token Retrieval
    var authUrl =
'https://mcgmpd574...auth.marketingcloudapis.com/v2/token';
    var payload = {
        "grant_type": "client_credentials",
        "client_id": "90ucil64n7dljpodm06zhcw",
        "client_secret": "o74qCBtGEYNrGp0frPL108Rp",
        "account_id": "534003610"
    };

    var tokenResponse = HTTP.Post(authUrl, 'application/json',
Stringify(payload));
    if (tokenResponse.StatusCode == 200) {
        var json = Platform.Function.ParseJSON(tokenResponse.Response[0]);
        var accessToken = json.access_token;
        var restUrl = json.rest_instance_url;

        // Call-2: Loop & Refresh FDE Objects
        for (var i = 0; i < objectKeys.length; i++) {
            var refreshUrl = restUrl + '/email/v1/filteredCustomObjects/' +
objectKeys[i] + '/refresh';
            var refreshRes = HTTP.Post(refreshUrl, 'application/json', '{}',
["Authorization"], ["Bearer " + accessToken]);
            Write("Refreshed: " + objectKeys[i] + " -> Status Code: " +
refreshRes.StatusCode + "<br>");
        }
    }
}
</script>
```



# Transition to Live Demo

Moving over to our development environment. We are going to trigger a filtered refresh in real-time, watching the data-sync live on screen.

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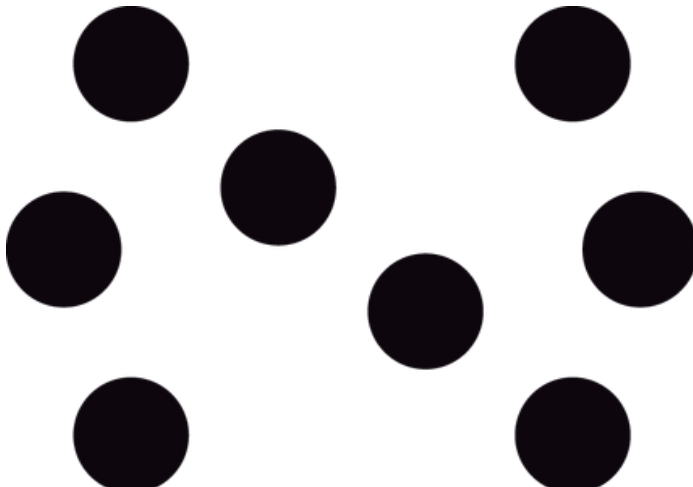
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# Thank You!

**Fnu Avi**

Senior Software Engineer,  
University of Southern  
California (USC)



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## Questions?

